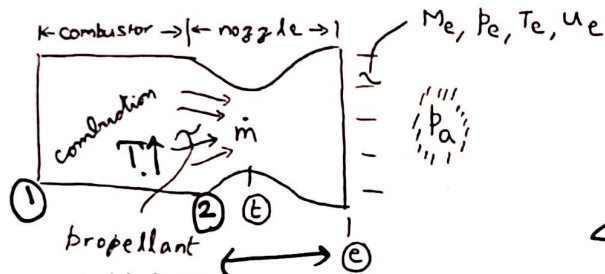


$p = cte$  in thrust chamber  
 $T = cte$  in nozzle  $\rightarrow h_{02} = h_{0e}$

$$\tau_2 \dot{m} u_e + \Delta p A$$

### Chemical Rocket Thrust chamber (ch 11)

\* consists of the combustion chamber and nozzle.



Put  $T_{02}$  f.r.m.  $\overline{p_0} = cte$

$$u_e = \sqrt{2c_p \left( T_{01} + \frac{Q_R}{c_p} \right) \left[ 1 - \left( \frac{p_e}{p_{02}} \right)^{(\gamma-1)/\gamma} \right]} \approx \sqrt{2Q_R \left[ 1 - \left( \frac{p_e}{p_{02}} \right)^{(\gamma-1)/\gamma} \right]} \quad (11.4)$$

$$\dot{m} = \frac{A^* p_{02}}{\sqrt{RT_{02}}} \sqrt{\gamma \left( \frac{2}{\gamma+1} \right)^{(\gamma+1)/(\gamma-1)}}$$

$$\mathcal{T} = \dot{m} u_e + (p_e - p_a) A_e$$

Eqs. (11.3) and (11.6)  
 $u_e$   $\dot{m}$

$$\dot{Q}_R = \dot{m}(h_{02} - h_{01}) = \dot{m}c_p(T_{02} - T_{01})$$

$$T_{02} = T_{01} + \frac{Q_R}{c_p} \quad (*)$$

$$h_{02} = h_{0e} \text{ or } \frac{u_e^2}{2} = h_{02} - h_e = c_p(T_{02} - T_e) \rightarrow = c_p T_{02} \left( 1 - \frac{T_e}{T_{02}} \right)$$

$$u_e = \sqrt{2c_p T_{02} \left[ 1 - \left( \frac{p_e}{p_{02}} \right)^{(\gamma-1)/\gamma} \right]}$$

$$u_e = \sqrt{\frac{2\gamma\bar{R}}{(\gamma-1)M} T_{02} \left[ 1 - \left( \frac{p_e}{p_{02}} \right)^{(\gamma-1)/\gamma} \right]} \quad (11.3)$$

$$\gamma = \frac{\delta\bar{R}}{(\delta-1)\bar{H}}$$

$$u_e \approx \sqrt{2 \frac{Q_R}{\dot{M}} \left[ 1 - \left( \frac{p_e}{p_{02}} \right)^{(\gamma-1)/\gamma} \right]} \quad (11.5)$$

$$Q_R = \frac{\bar{Q}_R}{\bar{M}}$$

Verify 11.7

H.W (3)

$$\frac{\mathcal{T}}{A^* p_0} = \sqrt{\frac{2\gamma^2}{\gamma-1} \left( \frac{2}{\gamma+1} \right)^{(\gamma+1)/(\gamma-1)} \left[ 1 - \left( \frac{p_e}{p_0} \right)^{(\gamma-1)/\gamma} \right]} + \left( \frac{p_e}{p_0} - \frac{p_a}{p_0} \right) \frac{A_e}{A^*} \quad (11.7)$$

$\dot{m} = f(M)$  put  $M=1 \rightarrow \dot{m} = f(\text{other})$   
 other

$$C_T = \frac{\mathcal{T}}{A^* p_0}$$

$$U_e = \sqrt{2C_p T_{02} \left[ 1 - \left( \frac{p_e}{p_{02}} \right)^{\frac{(\gamma-1)}{\gamma}} \right]}$$

$$U_e = \sqrt{\frac{2\gamma\bar{R}}{(\gamma-1)M} T_{02} \left[ 1 - \left( \frac{p_e}{p_{02}} \right)^{\frac{(\gamma-1)}{\gamma}} \right]}$$

$$U_e \approx \sqrt{2Q_R \left[ 1 - \left( \frac{p_e}{p_{02}} \right)^{\frac{(\gamma-1)}{\gamma}} \right]} = \sqrt{2 \frac{\bar{Q}_R}{\bar{M}} \left[ 1 - \left( \frac{p_e}{p_{02}} \right)^{\frac{(\gamma-1)}{\gamma}} \right]}$$

$$\uparrow \quad \uparrow \\ I_{sp} = u_e / g_e$$

- For high  $U_e$  (high  $I_{sp}$ ), desire
  - Propellants with low molecular weight,  $\bar{M}$
  - Propellant mixtures with large heat release,  $\bar{Q}_R$
  - High combustion chamber pressure,  $\underline{\underline{p_{02}}}$

$$\frac{p_e}{p_{02}} \downarrow \downarrow$$

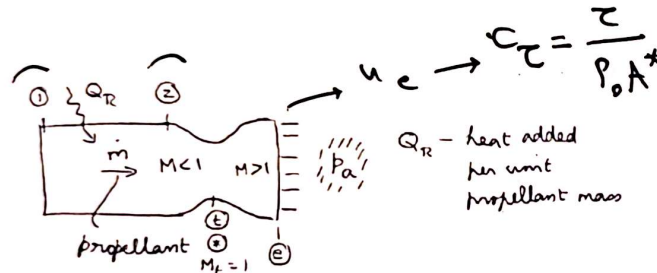
NOTE: Sometimes subscript 2 is dropped, but still conditions in combustion chamber

## Combustion Chamber

characteristic velocity  $c^* \stackrel{\text{def}}{=} \frac{p_0 A_t}{\dot{m}} \rightarrow p_0 A_t = \dot{m} c^*$

Thrust coefficient  $C_T \stackrel{\text{def}}{=} \frac{T}{p_0 A_t}$

$T = C_T (p_0 A_t) = (\dot{m} c^*) C_T, \quad T = \dot{m} c^* C_T$



I say across combustor:

$$\dot{m} c_p T_{01} + \dot{m} Q_R = \dot{m} c_p T_{02}$$

$$\Rightarrow T_{02} = T_{01} + \frac{Q_R}{c_p} \quad (11.1)$$

$$p_{02} = p_{01}$$

$T_{02}$  and  $p_{02}$  are referred to as chamber conditions, and are often represented as just  $T_0$  and  $p_0$ .  $T_{02}$  is also known as the flame temperature.

$$T_{02} \sim \frac{Q_R}{c_p} \text{ As } \frac{Q_R}{c_p} \gg T_{01}$$

\* Model (ideal rocket thrust chamber)

① Working fluid (propellant) is a perfect gas  $p = \rho R T$  with constant composition.

② Combustion is equivalent to ideal, constant-pressure heat addition.  $p_0$  constant ( $M=0$ )

③ Nozzle expansion is steady, one-dimensional and isentropic. recall Rayleigh-line result (from AE 308)

$$\frac{A}{A^*} = f(M)$$

The Mach number of the flow at the nozzle exit is found from either of the following stipulations:

① If nozzle area ratio is specified, this will determine  $M_e$ :

$$\frac{A_e}{A_t} \underset{M_e=1}{=} \frac{A_e}{A^*} = \frac{1}{M_e} \left[ \frac{2}{\gamma+1} \left( 1 + \frac{\gamma-1}{2} M_e^2 \right) \right]^{\frac{\gamma+1}{2(\gamma-1)}}$$

Select supersonic  $M$

② If the jet pressure at nozzle exit is known, this will determine  $M_e$ .

$$\frac{P_{01}}{P_e} = \frac{P_{02}}{P_e} = \frac{P_{0e}}{P_e} = \left(1 + \frac{\gamma-1}{2} M_e^2\right)^{\frac{\gamma}{\gamma-1}}$$

For isentropic nozzle (third assumption in the model),

$$T_{0e} = T_{02} \text{ and } P_{0e} = P_{02}$$

I law across nozzle:  $h_{0e} = h_{02} + \frac{u_e^2}{2}$

$$(or) h_e + \frac{u_e^2}{2} = h_{02} \Rightarrow c_p T_e + \frac{u_e^2}{2} = c_p T_{02}$$

$$(or) u_e = \sqrt{2 c_p (T_{02} - T_e)} \quad \left\{ T_e = \frac{T_{02}}{1 + \frac{\gamma-1}{2} M_e^2} \right\}$$

if I give you  $T_{02}$  &  $T_e \rightarrow$  you can get  $u_e$  &  $M_e$

Comments

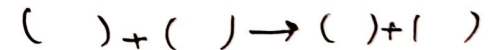
$$① \quad c_p = \frac{\gamma R}{\gamma-1} = \frac{\gamma}{\gamma-1} \frac{\bar{R}}{\bar{M}} \Rightarrow u_e = \sqrt{\frac{2\gamma}{\gamma-1} \frac{\bar{R}}{\bar{M}} (T_{02} - T_e)}$$

Low molar mass ( $\bar{M}$ ) yields higher  $u_e$

$$② \quad T_{02} = T_{01} + \frac{Q_R}{c_p} = T_{01} + \frac{1}{c_p} \frac{\bar{Q}_R}{\bar{M}}$$

Higher  $\left(\frac{\bar{Q}_R}{\bar{M}}\right)$  yields higher  $T_{02}$ , and, hence, higher  $u_e$ .

Both  $\bar{Q}_R$  and  $\bar{M}$  depend on fuel-oxidizer ratio of the reactant mixture.



$$= T_{01} + \frac{Q_R}{c_p}$$

$$\dot{m} = \frac{A^* p_{02}}{\sqrt{RT_{02}}} \sqrt{\gamma \left( \frac{2}{\gamma+1} \right)^{\frac{\gamma+1}{\gamma-1}}}, \quad (11.6)$$

Characteristic velocity  $c^* \stackrel{\text{def}}{=} \frac{\hat{p}_0 \hat{A}_t}{\dot{m}}$

2nd proof requested  
with HW#3

For the ideal rocket thrust chamber model,

Verify

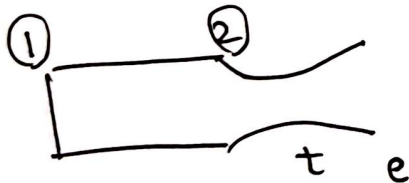
$$c^* = \sqrt{\frac{1}{\gamma} \left( \frac{\gamma+1}{2} \right)^{\frac{\gamma+1}{\gamma-1}} \frac{R T_{02}}{\bar{M}}} \quad (11.9)$$

{ recall  $T_{02}$  is the same as  $T_0$  }

$c^*$  is primarily dependent on combustion chamber properties (temperature, molar mass), and, therefore, it is a performance measure of the combustion part of the thrust chamber.

thrust coefficient  $C_T \stackrel{\text{def}}{=} \frac{T}{p_0 A_t}$

$$C_T = \sqrt{\frac{2\gamma^2}{\gamma-1} \left(\frac{2}{\gamma+1}\right)^{\frac{\gamma+1}{\gamma-1}}} \left[ 1 - \left(\frac{p_e}{p_0}\right)^{\frac{\gamma-1}{\gamma}} \right] + \left(\frac{p_e}{p_0} - \frac{p_a}{p_0}\right) \frac{A_e}{A^*} \quad (11.7)$$



$$p_{01} = p_{02} = p_{0e}$$

For the ideal rocket thrust chamber model,

$$C_T = \sqrt{\frac{2\gamma^2}{\gamma-1} \left(\frac{2}{\gamma+1}\right)^{\frac{\gamma+1}{\gamma-1}}} \left[ 1 - \left(\frac{p_e}{p_{02}}\right)^{\frac{\gamma-1}{\gamma}} \right] + \frac{p_e - p_a}{p_{02}} \frac{A_e}{A^*} \quad (11.11)$$

Put  $\frac{p_e}{p_0} = \left(\frac{2}{\gamma+1}\right)^{\frac{\gamma-1}{\gamma}}$  in 11.11

{ recall  $p_{02}$  is the same as  $p_0$  }

$C_T$  is dependent on nozzle characteristics and  $\Delta p$ , it is a measure of the performance of the nozzle part of the thrust chamber.

$(p_e - p_a)$

$$\frac{P_e}{P_0} = \left(1 + \frac{\gamma-1}{2} M_e^2\right)^{\frac{\gamma}{\gamma-1}} \quad \text{put } M_e=1 \rightarrow \frac{P_e}{P_0} = \left(1 + \frac{\gamma-1}{2}\right)^{\frac{\gamma}{\gamma-1}} = \left(\frac{\gamma+1}{2}\right)^{\frac{\gamma}{\gamma-1}}$$

If the nozzle does not have a diverging portion, then the pressure ratio  $p_e/p_0$  given by Eq. (3.11) with  $M = 1$ :

$$M=1 \rightarrow p_e/p_0 = \left(\frac{2}{\gamma+1}\right)^{\frac{\gamma}{\gamma-1}},$$

and the thrust coefficient reduces to

$$C_{g_{\text{conv}}} = (\gamma+1) \left(\frac{2}{\gamma+1}\right)^{\frac{\gamma}{\gamma-1}} - \frac{P_a}{P_0}, \quad (11.12)$$

which  $C_{g_{\text{conv}}}$  is the thrust coefficient for a converging nozzle. The ratio of thrusts, with and without a diverging section, is given by

$$\frac{\mathcal{T}}{\mathcal{T}_{\text{conv}}} = \frac{C_g}{C_{g_{\text{conv}}}} \rightarrow \frac{11.11}{11.12} \rightarrow \frac{\tau_{c-D}}{\tau_c}$$

and may be evaluated by means of Eqs. (11.11) and (11.12) above as a function of  $\gamma$ ,  $p_e/p_0$ , and  $p_a/p_0$ . For the special case  $\gamma = 1.2$ , Fig. 11.3 shows how the diverging portion of the nozzle contributes to the rocket thrust at various area and pressure ratios. We can see that for each value of the ambient pressure ratio  $p_a/p_0$  there is an optimum area ratio for the nozzle; it is the one for which  $p_e = p_a$  (see discussion of Fig. 10.2).

We can see from Fig. 11.3 that the thrust of a rocket will actually be reduced by adding a diverging section of too large an area ratio. However for reasonable pressure ratios ( $p_a/p_0 < 0.02$ ) this would not happen unless  $A_e/A^*$  were greater than about 30. For very small pressure ratios (such as  $p_a/p_0 = 0.001$ ), we will see that very large area ratios are associated with maximum thrust.

$$C_g = \frac{2\gamma^2}{\gamma-1} \left(\frac{2}{\gamma+1}\right)^{\frac{\gamma}{\gamma-1}} \left[1 - \left(\frac{p_e}{p_0}\right)^{\frac{\gamma-1}{\gamma}}\right] + \frac{p_e - p_a}{p_0} \frac{A_e}{A^*} \quad (11.11)$$

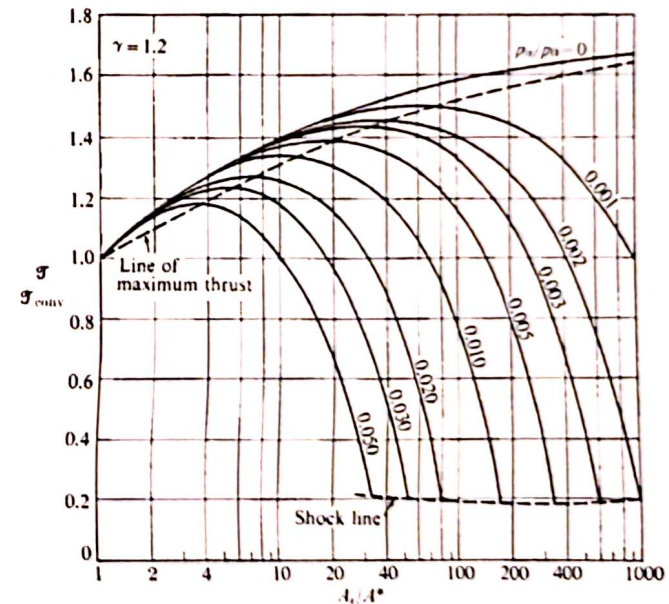
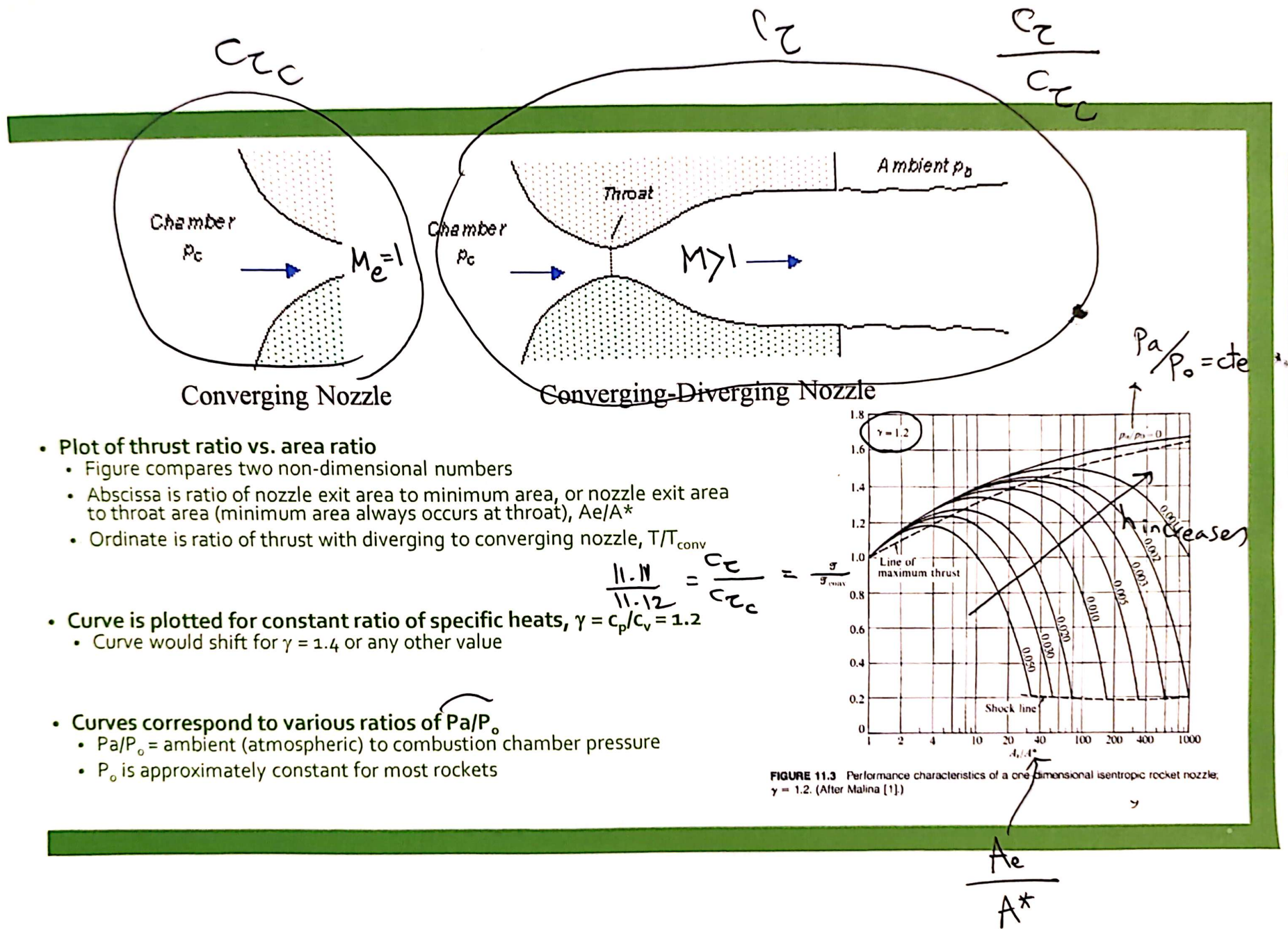


FIGURE 11.3 Performance characteristics of a one-dimensional isentropic rocket nozzle;  $\gamma = 1.2$ . (After Malina [1].)



- **Plot of thrust ratio vs. area ratio**

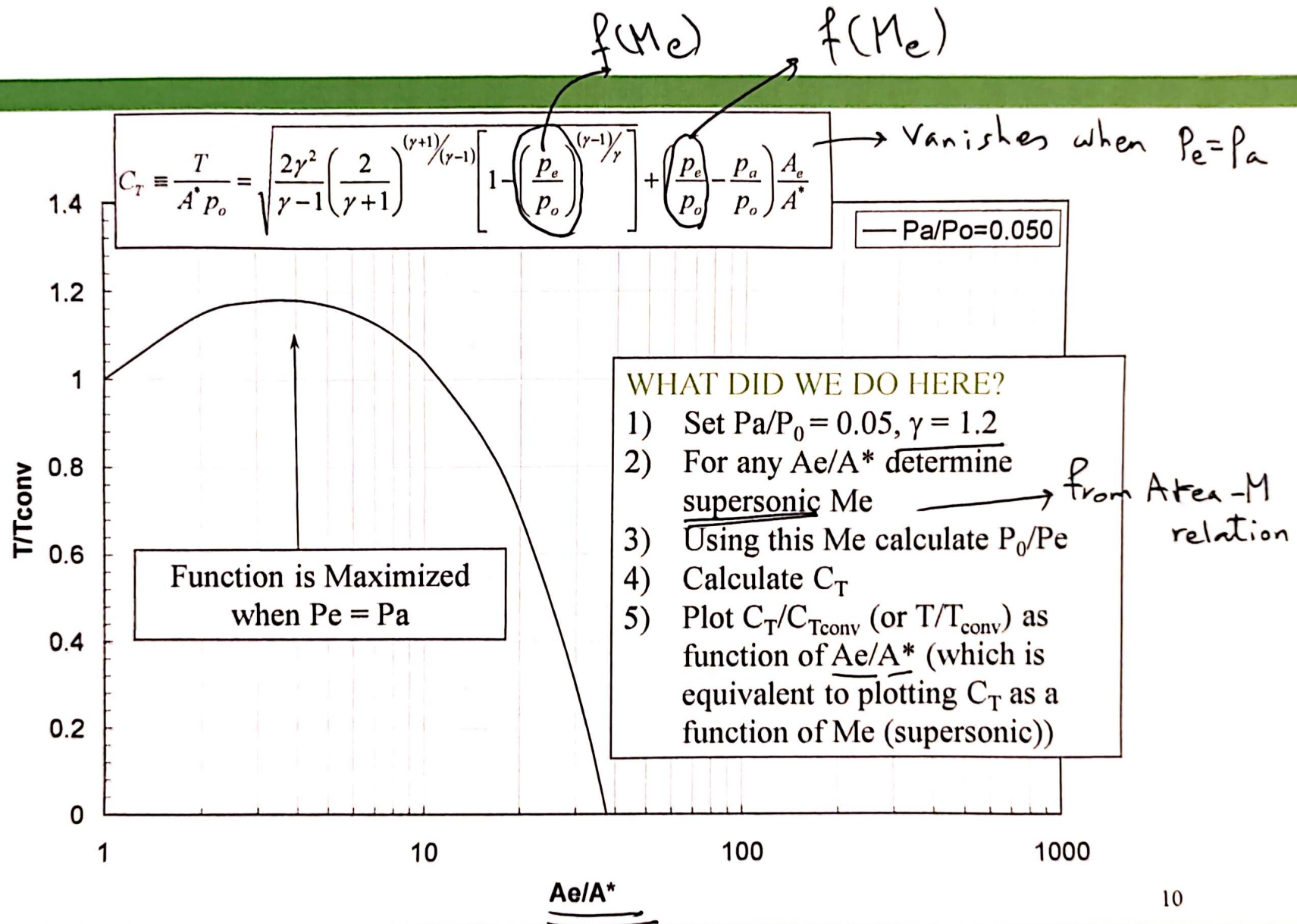
- Figure compares two non-dimensional numbers
- Abscissa is ratio of nozzle exit area to minimum area, or nozzle exit area to throat area (minimum area always occurs at throat),  $A_e/A^*$
- Ordinate is ratio of thrust with diverging to converging nozzle,  $T/T_{conv}$

$$\frac{11.11}{11.12} = \frac{c_{\tau}}{c_{\tau c}} = \frac{g}{g_{conv}}$$

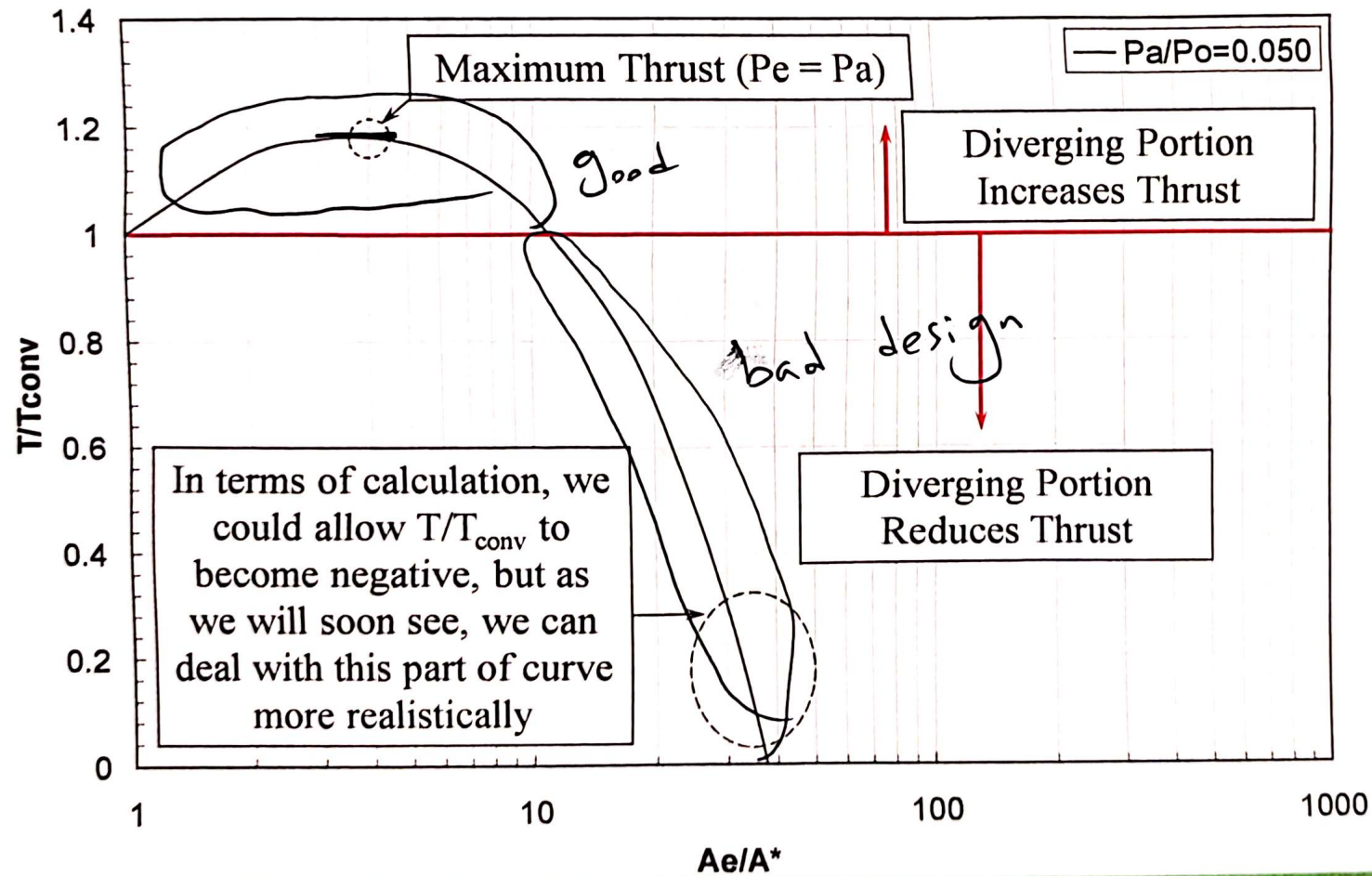
- **Curve is plotted for constant ratio of specific heats,  $\gamma = c_p/c_v = 1.2$**
- Curve would shift for  $\gamma = 1.4$  or any other value

- **Curves correspond to various ratios of  $P_a/P_0$**

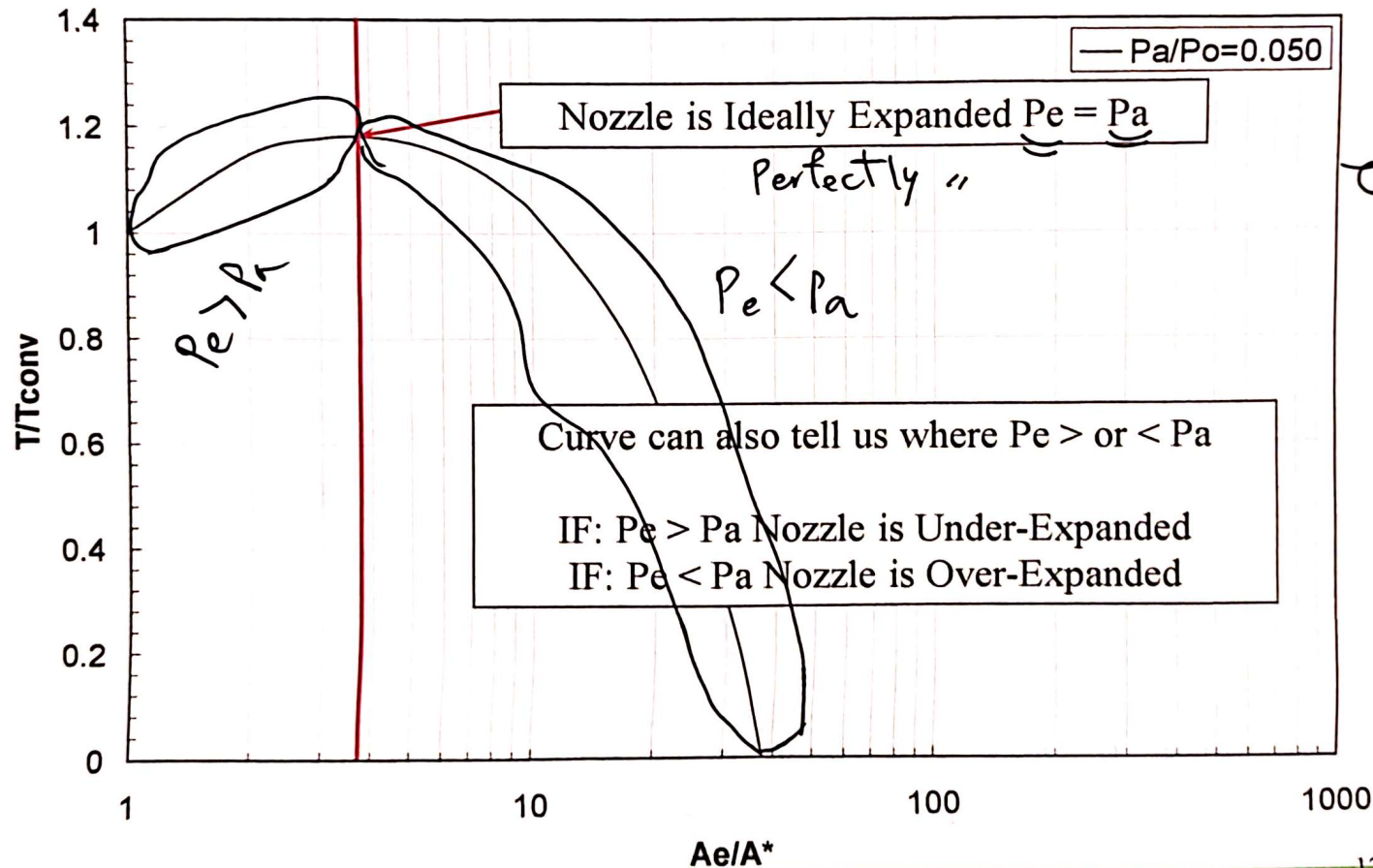
- $P_a/P_0$  = ambient (atmospheric) to combustion chamber pressure
- $P_0$  is approximately constant for most rockets



## PERFORMANCE CHARACTERISTIC OF A 1-D ISENTROPIC NOZZLE



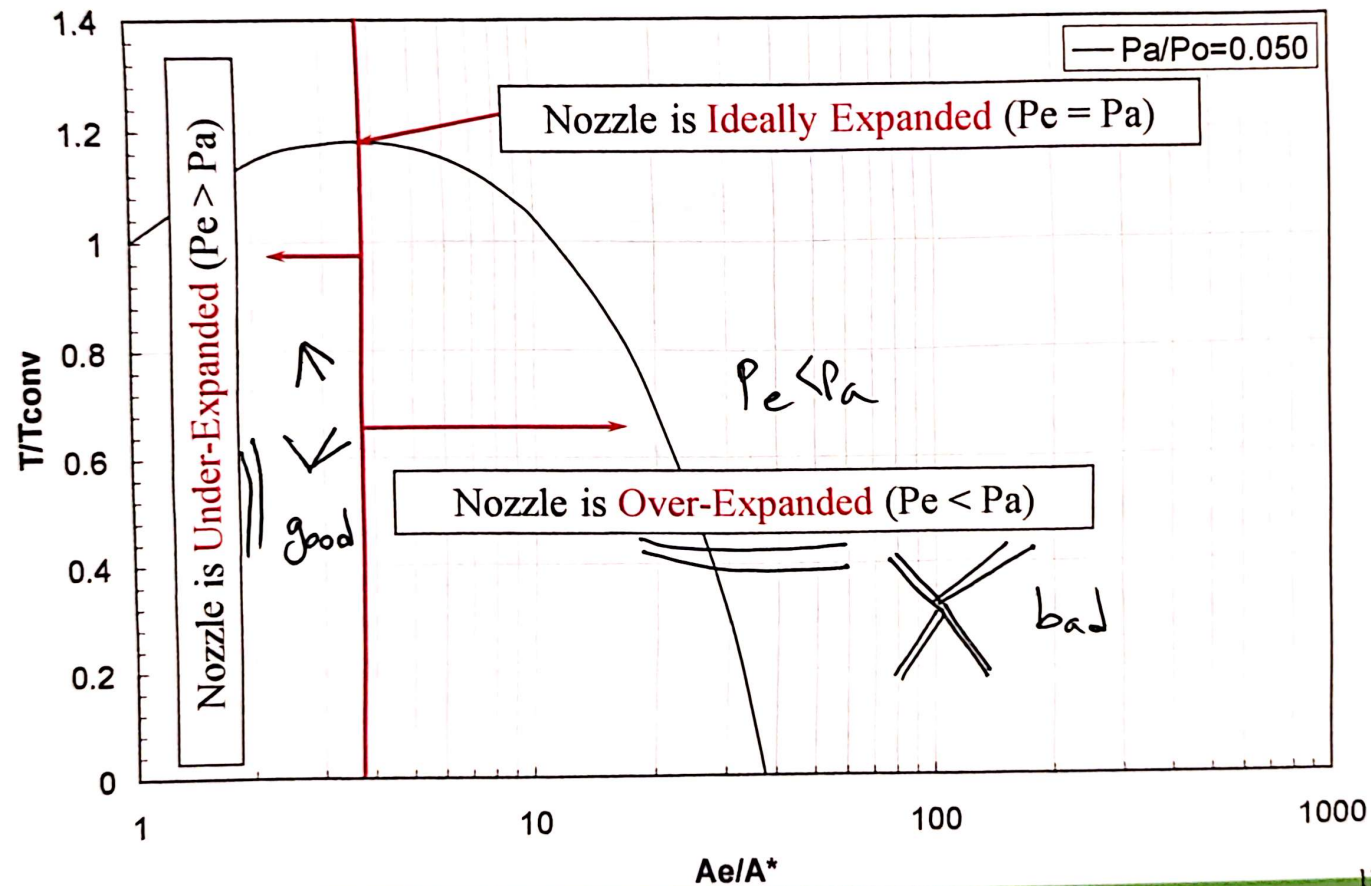
# PERFORMANCE CHARACTERISTIC OF A 1-D ISENTROPIC NOZZLE



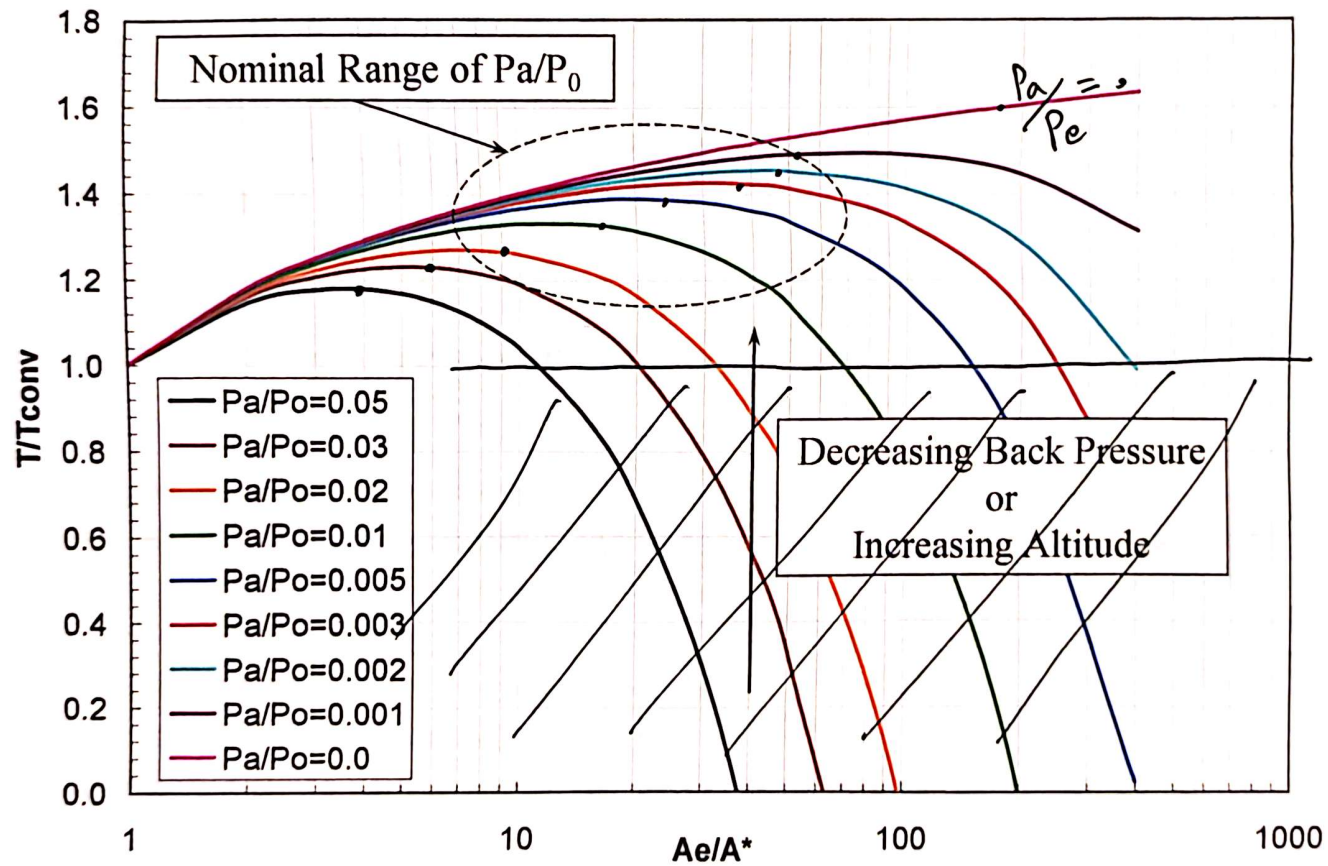
$$C = m u_e + \frac{(P_e - P_a) A_e}{\rho_e u_e}$$

< 0 when  $P_e < P_a$

## PERFORMANCE CHARACTERISTIC OF A 1-D ISENTROPIC NOZZLE



## PERFORMANCE CHARACTERISTIC OF A 1-D ISENTROPIC NOZZLE



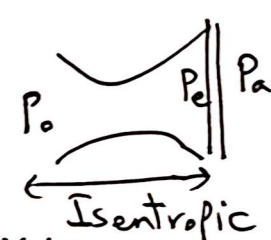
## KEY POINTS ON PERFORMANCE CURVE

- How does a rocket flying vertically move on Performance Curve?
  - High  $\underline{P_a/P_o}$  to Low  $\underline{P_a/P_o}$ 
    - $P_o$  usually remains ~ constant during flight  $P_o \sim cte$
    - $P_a \downarrow$  as altitude  $\uparrow$
- As  $P_a/P_o \downarrow$  very large  $A_e/A^*$  for maximum thrust
- How does optimal  $A_e/A^*$  vary as rocket flies vertically?
  - Required  $A_e/A^*$  for maximum thrust increases as rocket altitude increases
- If  $T/T_{conv} < 1$ , diverging portion of rocket is hindrance
  - Actual rockets never operate in this region
  - Best nozzle gives best performance ( $I_{sp}$ , range, etc.) over flight envelope

Figure 11.3 displays a shock line that shows the pressure ratios and area ratios for which a shock enters the nozzle. One may derive it as follows: Suppose that a normal shock sits in the exit plane of the nozzle and that the conditions just ahead of it are  $p_e$ ,  $M_e$ , and so on; the pressure just behind it is  $p_a$ . Then, according to Eq. (3.36) for normal shocks,

$$\frac{p_a}{p_e} = \frac{2\gamma}{\gamma+1} M_e^2 - \frac{\gamma-1}{\gamma+1}$$

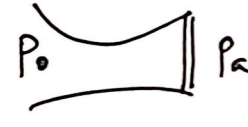
N.S. relation



For the isentropic nozzle the pressure ratio  $p_0/p_e$  is related to  $M_e$  by

$$\frac{p_0}{p_e} = \left(1 + \frac{\gamma-1}{2} M_e^2\right)^{\gamma(\gamma-1)}$$

so that

$$\frac{p_a}{p_0} = \left(\frac{2\gamma}{\gamma+1} M_e^2 - \frac{\gamma-1}{\gamma+1}\right) / \left(1 + \frac{\gamma-1}{2} M_e^2\right)^{\gamma(\gamma-1)}$$


Further, the area ratio is related to  $M_e$  by (applying Eq. 3.15)

$$\frac{A_e}{A^*} = \frac{1}{M_e} \left[ \frac{2}{\gamma+1} \left(1 + \frac{\gamma-1}{2} M_e^2\right) \right]^{(\gamma+1)/(2(\gamma-1))}$$

These last two relations, then, show that for a given  $\gamma$  there is only one pressure ratio  $p_a/p_0$  for which a normal shock will locate in the plane of a nozzle of a given area ratio  $A/A^*$ . Further discussion of back pressure effects follows in Section 11.3.

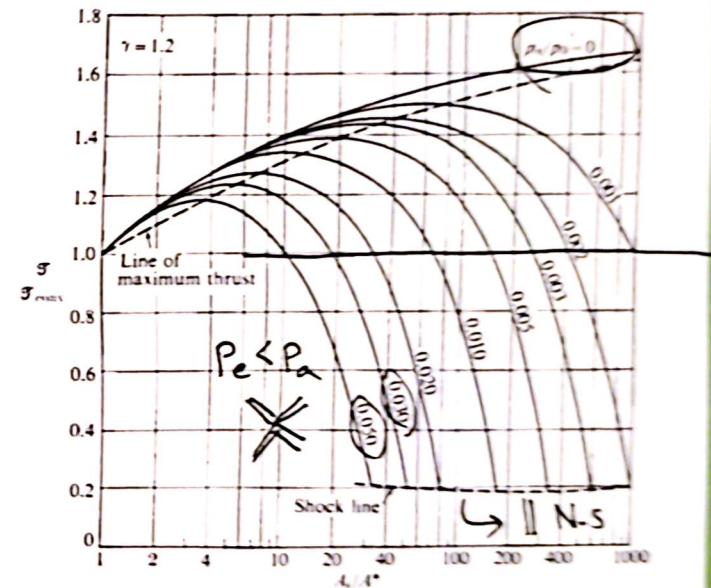


FIGURE 11.3 Performance characteristics of a one-dimensional isentropic rocket nozzle.  $\gamma = 1.2$ . (After Malina [1].)

$$\frac{p_a}{p_0} \checkmark \rightarrow \text{obtain } M_e \rightarrow \text{obtain } \frac{A_e}{A^*}$$

related to  
N.S